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Paper 1: Integrative QSP Modeling of Cancer Stem Cells in Triple-Negative Breast Cancer:

Bibliographic Information

- **Title:** Integrative QSP Modeling of Cancer Stem Cells in Triple-Negative Breast Cancer: Impacts on Tumor Immunity and Response to Immune Checkpoint Inhibitors
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Paper Type: POPK_PKPD

Quick Take

This study develops a detailed QSP model integrating cancer stem cell (CSC) biology with tumor immunity in triple-negative breast cancer. It quantitatively demonstrates how CSC properties like self-renewal, de-differentiation, and immunosuppressive cytokine secretion drive tumor progression, suppress immunity, and confer resistance to immune checkpoint inhibitors. Pharmacometricians involved in immuno-oncology will find this a robust framework for virtual trial design and combination therapy exploration.

Executive Summary

Xue et al. present a comprehensive QSP model for triple-negative breast cancer (TNBC) that mechanistically distinguishes cancer stem cells (CSCs) from differentiated tumor cells (DTCs) and integrates their interactions with the innate and adaptive immune system. Calibrated and validated using novel longitudinal murine data from wild-type and PD-L1 knockout 4T1 tumors, the model captures the dynamics of tumor volume, CSC frequency, and immune cell populations. Key findings show that enhanced CSC self-renewal (k_{ngCSC}), de-differentiation capacity (R_{de}), and cytokine secretion (TGF- β , VEGF) are the dominant drivers of immunosuppression and poor ICI response. The work bridges to clinical relevance by correlating in silico findings with TCGA TNBC transcriptomic data, establishing CSC frequency as a potential predictive biomarker and providing a quantitative framework for developing anti-CSC/ICI combination strategies.

Scientific Context & Motivation

Low response rates to ICIs in TNBC are a major clinical challenge, and CSCs are hypothesized to be key mediators of resistance. However, prior mechanistic studies have been fragmented, focusing on single pathways without an integrative quantitative assessment. Existing mathematical models lack granularity in distinguishing CSCs from DTCs or suffer from a lack of longitudinal calibration data, particularly for CSC and immune cell dynamics. This study addresses a critical gap by

building a multi-mechanistic, data-calibrated QSP model that simultaneously evaluates how CSC-intrinsic properties (proliferation, de-differentiation, cytokine secretion, checkpoint expression) shape the tumor microenvironment and dictate therapeutic outcomes, providing a systems-level understanding that single-mechanism studies cannot.

✂ Methodological Snapshot

A large-scale, mechanism-based QSP model built in MATLAB SimBiology integrates the cancer-immune cycle with a novel, detailed representation of CSCs. It differentiates CSCs from DTCs across 8 distinct biological features, including cell volume, proliferation, resistance to immune killing, checkpoint expression, and cytokine secretion. The model is calibrated using a multi-step workflow against unique, unified longitudinal data from murine TNBC models (4T1 and 4T1-PDL1KO) capturing tumor size, CSC frequency, and innate/adaptive immune cell dynamics. LHS-PRCC sensitivity analysis and virtual patient simulations are then used to interrogate the model and bridge findings to clinical TCGA data analysis.

☒ Structural Model Breakdown

The model comprises five compartments: Tumor (T), Blood Vessel (BV), Lymph Node (LN), Lymph Transport System (LTS), and Peripheral (P). Within the tumor compartment, the total tumor cell population is divided into CSCs and DTCs. Both populations can self-renew, with DTCs having a higher proliferation rate. CSCs differentiate into DTCs, and a fraction of DTCs de-differentiate back to CSCs, governed by a feedback mechanism to maintain equilibrium. The immune module includes detailed antigen processing by dendritic cells (DCs), activation and trafficking of CTLs, Tregs, Th1 cells, and NK cells, their tumor-killing functions, and the secretion of cytokines like TGF- β , VEGF, IFN- γ , and IL-2. The PD-1/PD-L1 axis is modeled with distinct binding and inhibitory effects for CSCs and DTCs. ICIs ($\alpha PD - 1$, $\alpha CTLA - 4$) are modeled with simple PK and their effect on inhibiting checkpoint-mediated suppression of T cell activity.

Detailed Methodological Analysis

Modeling Approach

A deterministic, compartmental ODE-based QSP model representing the tumor, blood vessel, lymph node, lymph transport system, and a peripheral compartment. The model incorporates 8 mechanistic distinctions between cancer stem cells (CSCs) and differentiated tumor cells (DTCs): relative cell volume, reduced CSC self-renewal rate (k_{ngCSC}), enhanced CSC resistance to NK and CTL killing, lower CSC MHC-I expression and IFN- γ inducibility, lower CSC immunogenicity, higher CSC expression of PD-L1, higher CSC secretion rates for TGF- β and VEGF, and stronger CSC-mediated CTL/NK inhibition per unit PD-1/PD-L1 complex. The model was built using MATLAB SimBiology R2022a.

Data Sources

All calibration and validation data were generated in-house using a BALB/c mouse model. Allografts were established from wild-type 4T1 cells, Cd274-knockout (PD-L1 KO) 4T1 cells, and their respective CSC-enriched populations derived from spheroid cultures. Longitudinal datasets collected include tumor volume (caliper), CSC frequency (ALDEFLUOR assay/flow cytometry), and intratumoral immune cell frequencies (flow cytometry including CTLs, Tregs, NK cells, DCs, TAMs). Clinical validation and bridging used RNA-seq data from 173 TNBC samples from the TCGA database, analyzed via CIBERSORTx for CTL infiltration and a literature-based 7-gene signature for stemness.

Estimation Methods

Parameters were calibrated using a manual stepwise approach. Literature-derived parameters were fixed, while CSC- and immune-related parameters were sequentially tuned. The process started by fitting innate immune cell and CSC frequencies in a PDL1-KO background to isolate CSC-intrinsic effects, then expanded to adaptive immunity and the full 4T1 model. The goodness-of-fit was visually assessed by comparing model simulations with observed mean and standard deviation of longitudinal data.

Model Evaluation

Model validation was performed by comparing simulations against an independent dataset: the growth and immune dynamics of tumors derived from inoculating 5×10^4 purified CSCs from 4T1 or 4T1-PDL1KO lines. The model’s clinical relevance was further evaluated by comparing its qualitative predictions (e.g., negative correlation between stemness and CTL infiltration) against an independent analysis of TCGA data using Spearman correlation.

Covariate Analysis

No formal covariate modeling was performed. Instead, sensitivity analysis using Latin Hypercube Sampling (LHS) and Partial Rank Correlation Coefficients (PRCC) was used to systematically identify which CSC-specific parameters (e.g., k_{ngCSC} , R_{dc} , secretion rates) most strongly influence tumor volume, immune landscape, and ICI efficacy, serving as a global sensitivity analysis of the model’s most critical mechanistic ‘covariates’.

Statistical Rigor Assessment

The use of LHS-PRCC for global sensitivity analysis is appropriate for this large, non-linear model and effectively identifies influential parameters without assumptions of normality or linearity. The virtual patient approach ($n = 2000$) provides a robust framework for exploring the parametric space and assessing outcome variability. The bridging of model predictions to clinical TCGA data using Spearman correlation is a pragmatic step for external validation. However, the model calibration itself is manual/visual, without a formal objective function value (OFV) or objective parameter uncertainty quantification (e.g., bootstrap, asymptotic confidence intervals), which limits the statistical assessment of parameter precision and model fit robustness. The choice of 60% TGI cutoff is pragmatic but arbitrary.

☒ Key Findings

1. The model successfully captured the divergent immune landscapes of 4T1 vs. 4T1-PDL1KO tumors, with PD-L1-deficient CSCs leading to increased CTL infiltration and slower growth.
 2. Enhanced CSC stemness (high k_{ngCSC} , R_{dc} , TGF- β /VEGF secretion) is a dominant driver of immune exclusion, creating a ‘cold tumor’ phenotype.
 3. CSC accumulation is primarily determined by tumor cell-intrinsic proliferation/de-differentiation dynamics, not by the immune microenvironment.
 4. High CSC stemness significantly impairs the efficacy of $\alpha PD-1$, $\alpha CTLA-4$, and their combination.
 5. Sensitivity analysis via LHS-PRCC and virtual patient simulations identified CSC self-renewal rate (k_{ngCSC}) and CSC frequency as top predictors of ICI response ($AUCs \approx 0.72$), outperforming DTC-related metrics.
 6. ICI administration sequence and timing critically modulate outcomes; sequential therapy ($\alpha PD-1$ followed by $\alpha CTLA-4$) and early intervention showed superior efficacy.
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🔗 Clinical & Regulatory Implications

The study provides strong quantitative support for evaluating tumor stemness (specifically CSC frequency) as a predictive biomarker for ICI response in TNBC, suggesting that baseline biopsy assessment could guide treatment decisions. It stresses that ICIs alone are insufficient in high-stemness tumors and makes a compelling argument for developing combination strategies that directly target CSC self-renewal (k_{ngCSC} , R_{dc}) alongside checkpoint blockade. The finding that CSC frequency can paradoxically increase with effective tumor killing highlights the risk of ICI-mediated CSC enrichment, further strengthening the case for upfront combination with anti-CSC agents. The work also provides a computational platform to preclinically optimize dosing sequence and timing for such combinations.

Strengths & Limitations

Strengths

- Highly granular mechanistic framework: The model uniquely distinguishes CSCs from DTCs across 8 distinct biological properties (volume, proliferation rate, immunogenicity, MHC-I expression, PD-L1 expression, cytokine secretion, resistance to killing, PD-1/PD-L1 inhibitory signaling), providing unparalleled detail.
- Robust multi-dimensional calibration: The model is calibrated and validated using a purpose-built, unified longitudinal dataset that simultaneously captures tumor volume, CSC frequency, and multi-parameter immune cell dynamics (CTLs, Tregs, NKs, DCs), a significant advance over models calibrated on tumor volume alone.
- Powerful dual sensitivity analysis (LHS-PRCC and virtual patients): This approach rigorously identifies the most influential CSC parameters and bridges them to clinically meaningful outcomes, establishing a hierarchy of mechanisms (k_{ngCSC} and $R_{dc} >$ cytokine secretion $>$ PD-L1 effects).

- Direct clinical bridging: The in silico findings are validated against TCGA data, demonstrating a consistent negative correlation between stemness scores and CTL infiltration, and establishing CSC frequency as a potential clinically measurable predictive biomarker.

Limitations (Acknowledged by Authors)

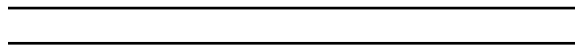
- Lack of spatial detail: The model uses a compartmental ODE approach and cannot capture spatial heterogeneity or CSC niche-dependent phenotypes. The authors themselves suggest agent-based models or PDEs for future work.
- Absence of stromal components: Cancer-associated fibroblasts (CAFs) and other stromal elements that influence CSCs and the microenvironment are not included, which the authors identify as a key area for model expansion.
- Complexity and potential for non-identifiability: The authors acknowledge that the high model complexity, while mechanistic, poses challenges for complete alignment with clinical data and may lead to unresolved parameter non-identifiability issues.

Limitations (Expert Review)

- All-murine calibration data: The model is entirely calibrated on murine 4T1 syngeneic models. While the authors justify cross-species parameter handling, the direct translatability to human TNBC biology, particularly CSC dynamics and cytokine signaling, remains a substantial assumption not formally tested.
- Oversimplified ICI PK/PD: ICI pharmacokinetics are modeled simply, and PD efficacy is calibrated to tumor growth curves without a detailed mechanistic breakdown of target engagement, receptor occupancy, or Fc-mediated effects, which can be critical for mAb therapies.
- CSC measurement proxy: The study uses a single murine model and a specific stemness score for clinical validation. CSC biomarkers are highly context-dependent and controversial; the finding that CSC frequency is a strong predictor may be specific to the model structure and chosen biomarkers, limiting immediate clinical deployability.

Generalizability

The model's core structure is built on fundamental cancer-immune cycle principles and could be adapted to other solid tumors with high stemness, such as pancreatic or glioblastoma. However, the quantitative parameter values and specific findings are heavily tailored to the 4T1 TNBC model and the specific murine BALB/c immune system, requiring re-calibration for other tumor types or clinical human settings. The qualitative insights on the dominance of CSC-intrinsic features for immunosuppression and ICI resistance are likely generalizable.



Figures & Tables

- **Figure 1:** Schematic overview of the QSP model. Panel A details the known biological mechanisms by which CSCs inhibit the cancer-immune cycle. Panel B maps these mechanisms onto the model's compartmental structure: Lymph Node (LN), Blood Vessel (BV), Tumor (T), Lymph Transport System (LTS), and Peripheral (P) compartments, showing the dynamic traffic of tumor cells, innate and adaptive immune cells, and cytokines.

- *Significance*: Provides the essential biological and mathematical blueprint for the entire study, clearly linking mechanistic hypotheses to model topology.
- **Figure 2**: Model calibration and validation workflow and results. Panel A shows the stepwise calibration strategy. Panels B-E display model predictions versus observed data for tumor volumes and percentages of key immune cells (NK, DC, TAM, CTL, Treg) and CSCs in various inoculation scenarios (4T1, 4T1-PDL1KO, CSCs enriched populations), the benchmark for model credibility.
 - *Significance*: This figure demonstrates the model’s strong and multi-faceted predictive performance against longitudinal experimental data.
- **Figure 3**: Simulated impact of varying initial CSC frequency (5%, 50%, 95%) on tumor volume, intratumoral immune cell densities, and cytokine levels over time.
 - *Significance*: Key visual representation showing that CSC frequency drives profound immune suppression but, due to differentiation equilibrium, does not drastically alter long-term tumor growth trajectory.
- **Figure 4**: Sensitivity analysis of CSC-specific parameters. Panel A is a PRCC heatmap linking CSC parameters (k_{ngCSC} , R_{dc} , $k_{secVEGFCSC}$, etc.) to tumor volume and immune cell metrics. Panel B illustrates the effect of completely ablating PD-L1 on CSCs alone. Panel C compares tumor growth and CTL density in ‘StemHigh’ vs ‘StemLow’ virtual tumors under different host immune strengths. Panel D is a clinical TCGA validation showing a significant negative correlation between stemness score and CTL score in human TNBC.
 - *Significance*: Identifies k_{ngCSC} , R_{dc} , and TGF- β /VEGF secretion as the dominant CSC drivers of tumor growth and immunosuppression, and confirms these predictions with real-world clinical transcriptomic data.
- **Figure 5**: Investigation of CSC accumulation dynamics. Simulates cold vs. hot tumors and compares tumor growth, immune infiltration, CSC frequency, and CSC number. Shows a new PRCC analysis identifying the parameters that dominantly control CSC frequency and number.
 - *Significance*: Reveals the critical insight that CSC accumulation is a tumor-intrinsic process driven primarily by proliferation/de-differentiation rates (k_{ngCSC} , R_{dc}), with limited influence from the immune microenvironment.
- **Figure 6**: Assessment of ICI efficacy and CSC-mediated resistance. Panel A shows calibration of $\alpha CTLA-4$ and $\alpha PD-1$ to tumor growth curves. Panels B-D compare TGI and CTL infiltration in StemHigh vs. StemLow tumors. Panel E is a PRCC heatmap of CSC parameters vs. TGI and post-treatment immune cell densities, highlighting the mechanistic drivers of ICI resistance.
 - *Significance*: Quantifies the severely compromised ICI efficacy in high-stemness tumors and highlights the distinct mechanistic underpinnings of resistance to $\alpha CTLA-4$ vs. $\alpha PD-1$.
- **Figure 7**: Identification of key determinants and biomarkers for ICI response. Panel A compares parameter values between ICI responders (TGI $\geq 60\%$) and non-responders. Panel B shows differential TGI when stratified by high/low k_{ngCSC} and R_{dc} . Panel C demonstrates that pre-treatment CSC frequency is a strong predictor of response. Panel D and E present ROC analyses showing that k_{ngCSC} and CSC frequency ($AUC \approx 0.72$) are the top univariate predictors of ICI response.

- *Significance*: Establishes CSC-related parameters and measurable CSC frequency as novel, potent predictive biomarkers for ICI outcomes, directly informing clinical patient stratification strategies.
 - **Figure S12**: Simulated impact of ICI administration sequence and treatment timing on TGI, CTL density, and CSC frequency.
 - *Significance*: Provides actionable insights into clinical trial design, showing that sequential $\alpha PD - 1 \rightarrow \alpha CTLA - 4$ therapy and early treatment initiation are superior strategies, and that effective tumor control paradoxically enriches for CSCs, underscoring the need for combination therapy.
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Supplementary Materials

The supporting information provides critical methodological details: (1) a comprehensive table (Table S1) defining the 8 key differences parameterized between CSCs and DTCs and the relative quantitative relationships used; (2) detailed methods for animal experiments, including derivation of allografts from CSC-enriched spheroids; (3) Tables S3-S5 defining parameter ranges for the ‘StemHigh/StemLow’, ‘Cold/Hot’, and virtual patient simulations; and (4) additional simulation results (Figures S3-S11) extending the main findings on PK, ICI efficacy under varying stemness and timing, and immune dynamics in responder vs. non-responder groups.

Future Directions

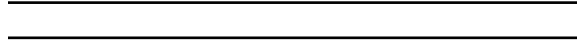
Key questions raised include: 1) Can the model be expanded with spatial components (PDE/agent-based) to simulate CSC niches and drug penetration? 2) What is the contribution of stromal cells like CAFs, and how are they modulated by CSC-derived TGF- β ? 3) Can this model be cross-calibrated with emerging clinical single-cell RNA-seq data to create a humanized platform? 4) Which specific anti-CSC targets (e.g., Notch, Wnt, CD44) should be prioritized for virtual combination trials? 5) Can a simplified, clinically identifiable sub-model be derived to enable patient-specific Bayesian forecasting based on baseline biopsy data?

Expert Commentary

This is a landmark study in immuno-oncology QSP, executed with a level of rigor that sets a new standard for the field. The comprehensive, eight-point distinction between CSC and DTC biology is a major conceptual and computational achievement. The workflow for multi-dimensional calibration, leveraging PDL1-KO cells as a validation tool, is exceptionally clever and robust. The pragmatic identification of k_{ngCSC} and R_{dc} as the dominant drivers is crucial—it tells us that while cytokines and checkpoints matter, the core proliferative ‘stemness engine’ is the ultimate arbiter of tumor aggression and ICI failure. A pearl from this work for junior modelers is the demonstration that CSC frequency is an intrinsic rather than microenvironmentally determined property, a counter-intuitive finding that profoundly impacts therapeutic strategy. The primary limitation is the species gap, but the model architecture itself is a major contribution that will undoubtedly be translated to human clinical data.

Bottom Line

For pharmacometricians, this QSP model provides a credible and detailed virtual platform to explore the complex interplay between cancer stemness and immunotherapy. It quantitatively confirms that CSC self-renewal and de-differentiation are the most potent levers controlling immune exclusion and ICI failure, advocating for combination strategies that directly target these CSC properties. The model serves as a foundation for virtual clinical trials to optimize dosing sequence and timing for anti-CSC and ICI combinations, with CSC frequency emerging as a key biomarker for patient stratification.



☒ Figures

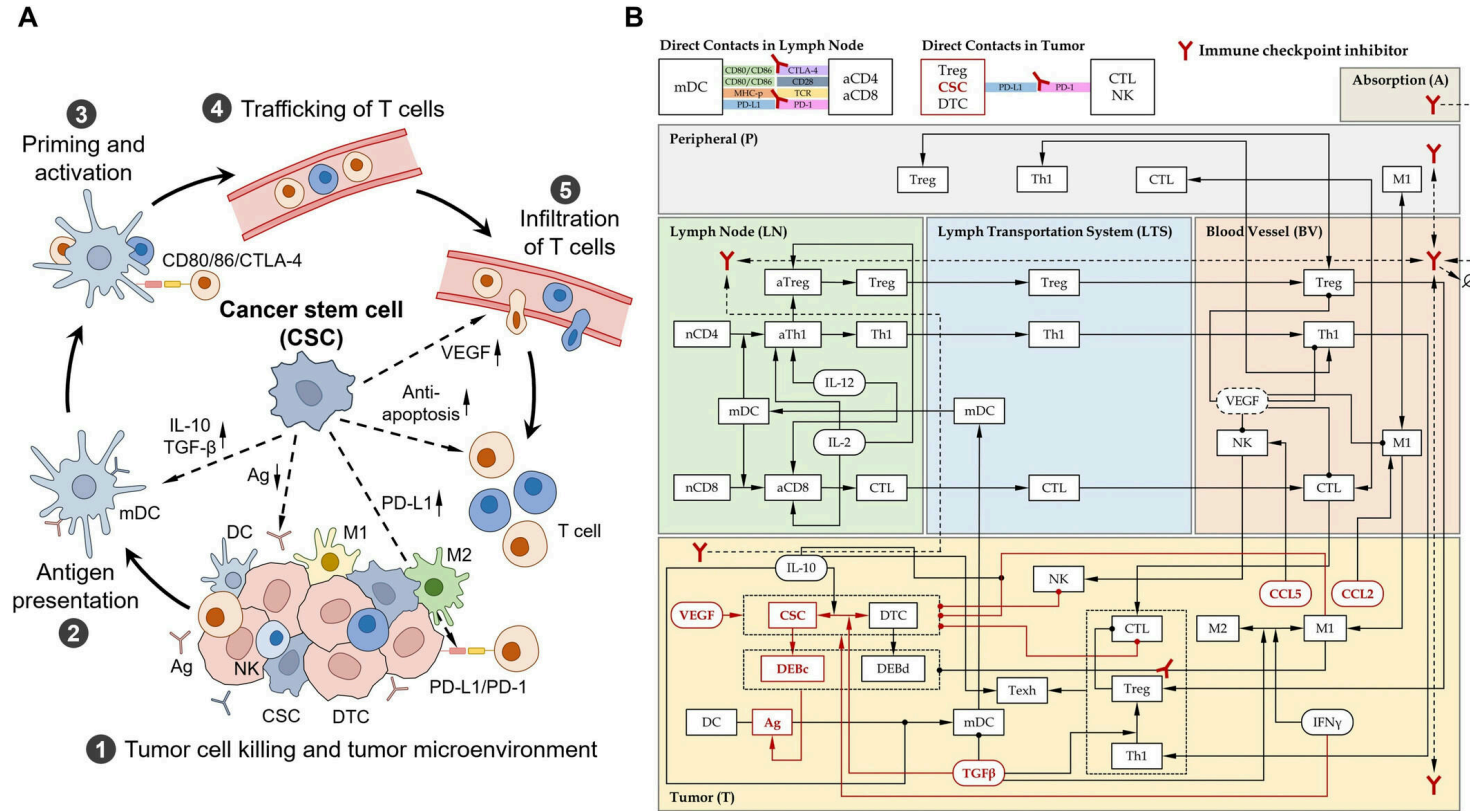


Figure 1: CSC immune mechanisms and QSP model. (A) Known mechanisms of CSCs hindering the cancer-immune cycle. (B) QSP model schematic diagram. Arrows indicate migration o

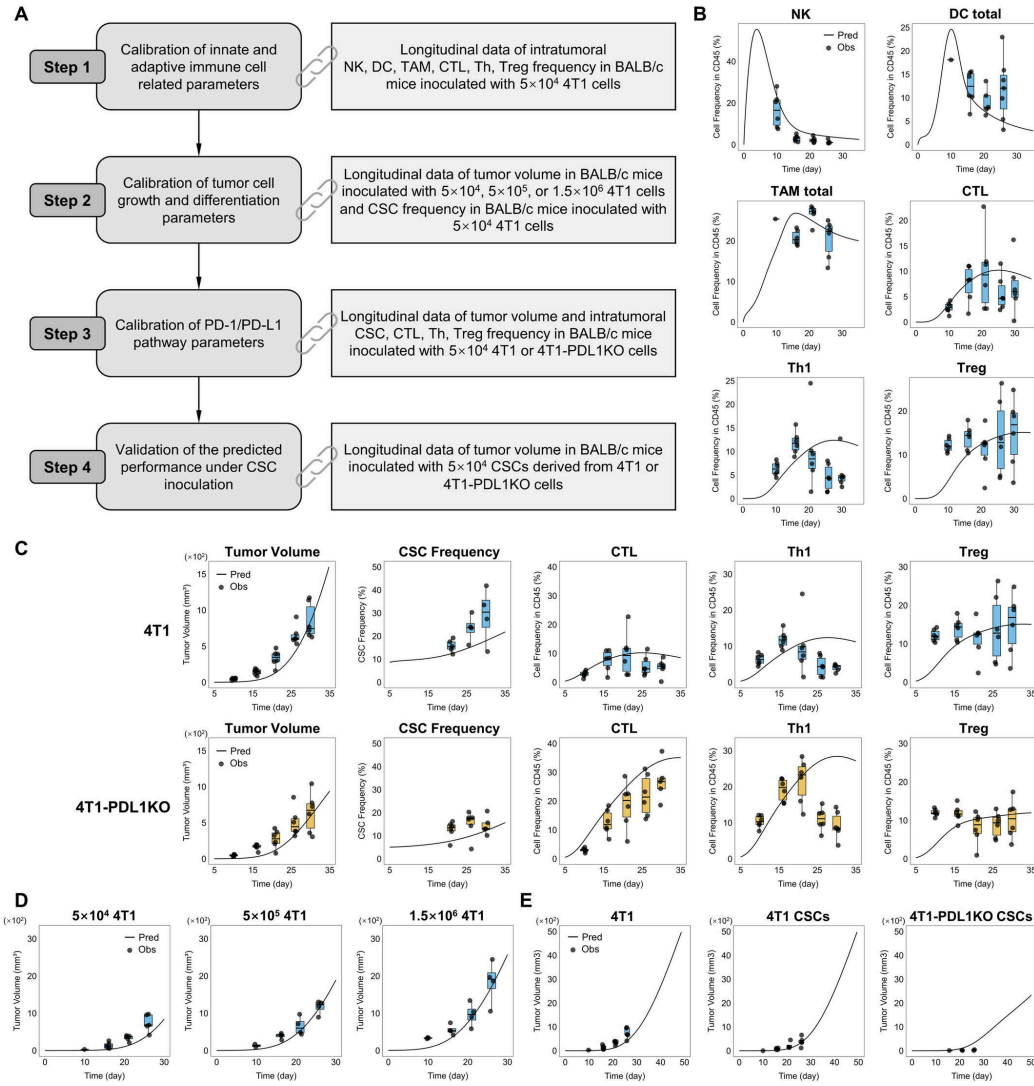


Figure 2: Calibration and validation of the QSP model. (A) Basic workflow of parameter calibration and validation. (B) Calibration step 1: The frequencies of NK, DC, TAM,

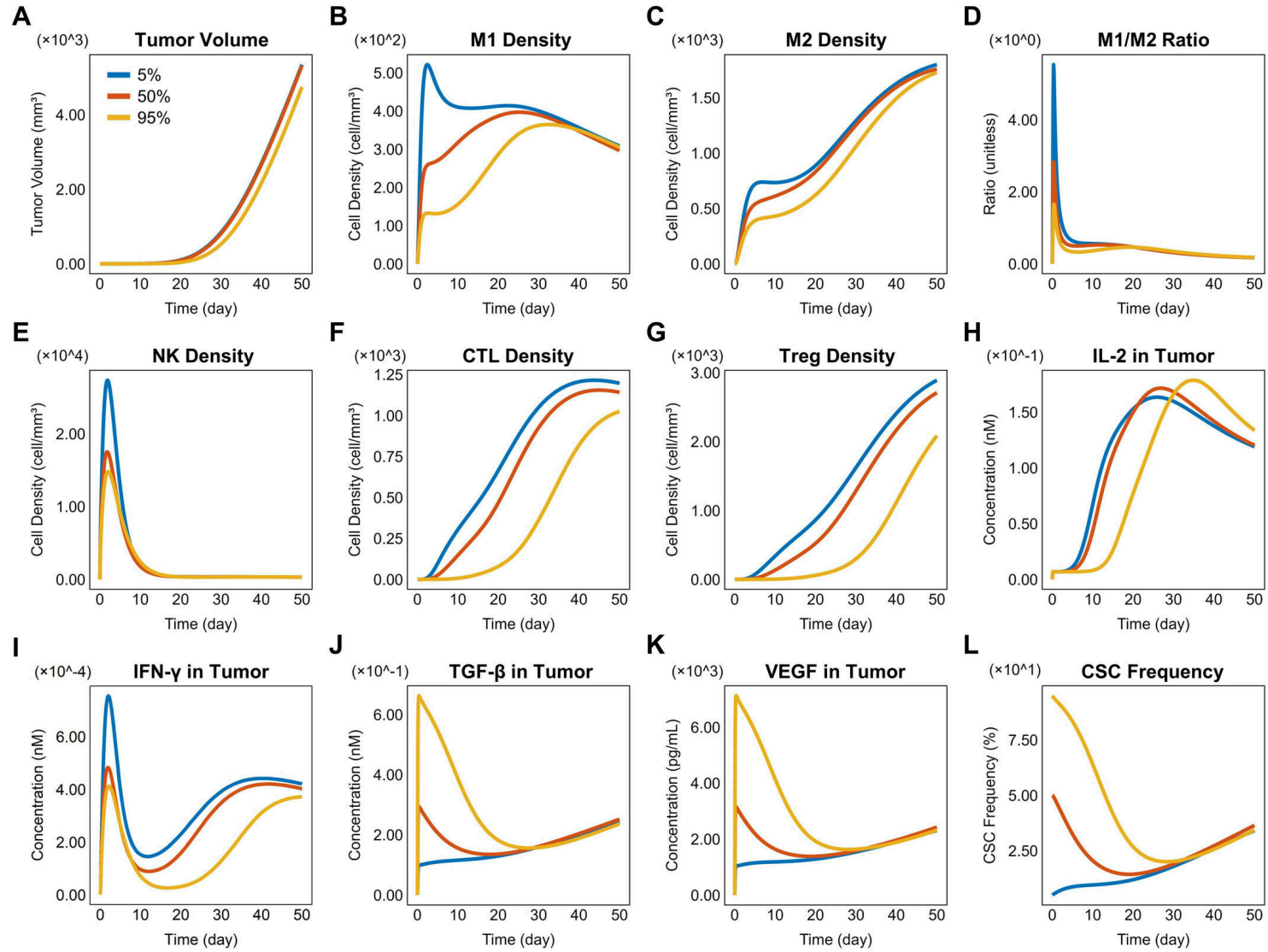


Figure 3: The simulated tumor volumes, intratumoral immune cells, and cytokines after inoculating 5×10^4 Tumor cells with initial CSC frequencies of 5%, 50%, and 95%. (A)

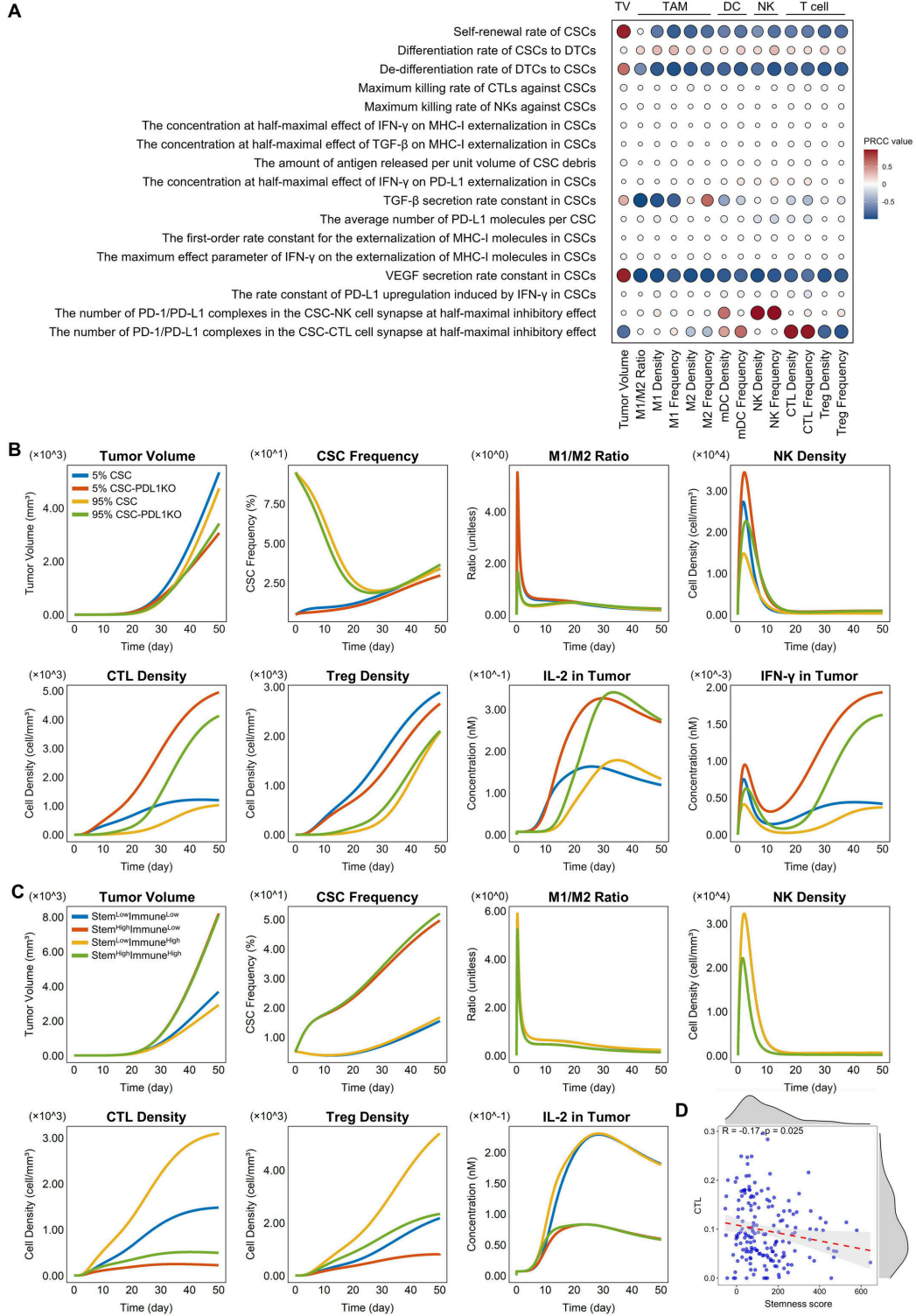


Figure 4: The key CSC characteristics that affect tumor growth and immunity. (A) The PRCC heatmap of CSC-specific parameters versus tumor volume, immune cell frequencies,

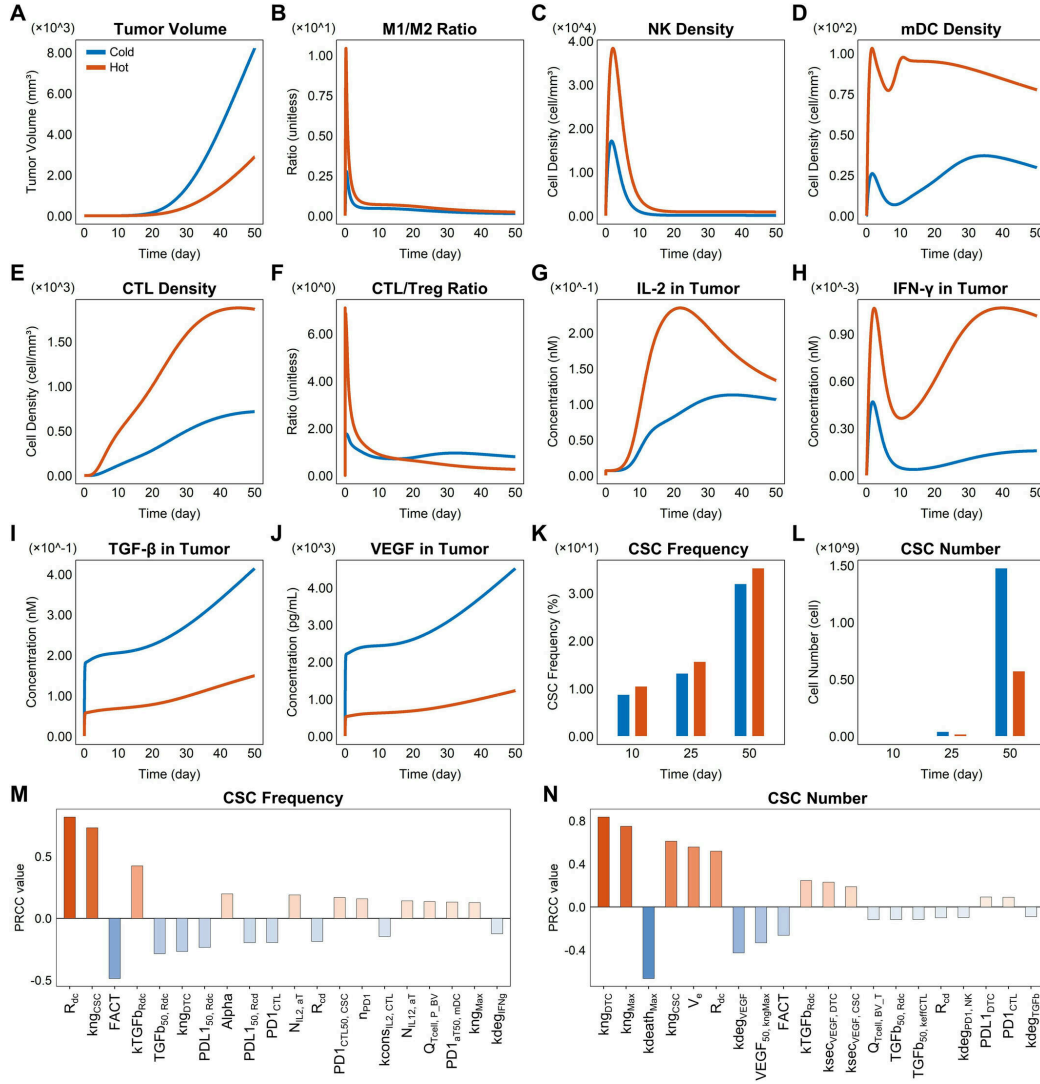


Figure 5: The simulated dynamic changes in tumor volumes, CSC frequencies, and immune indicators across cold and hot TNBC tumors, and the identification of dominant CSC-mo

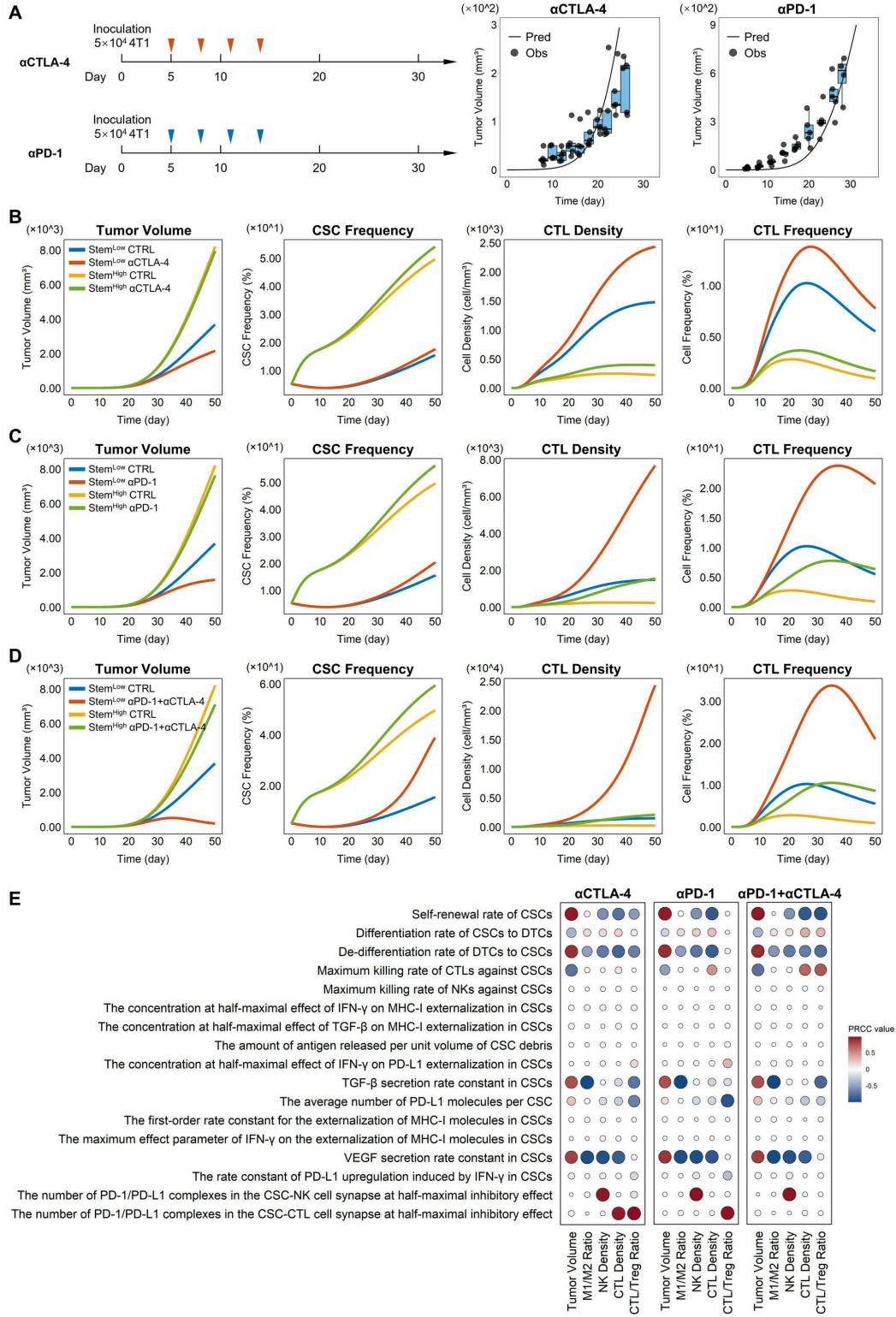


Figure 6: CSC-mediated impairment of ICI efficacy in TNBC and the underlying key characteristics. (A) Calibration of the efficacy parameter of α CTLA-4 or α PD-1 using tumor

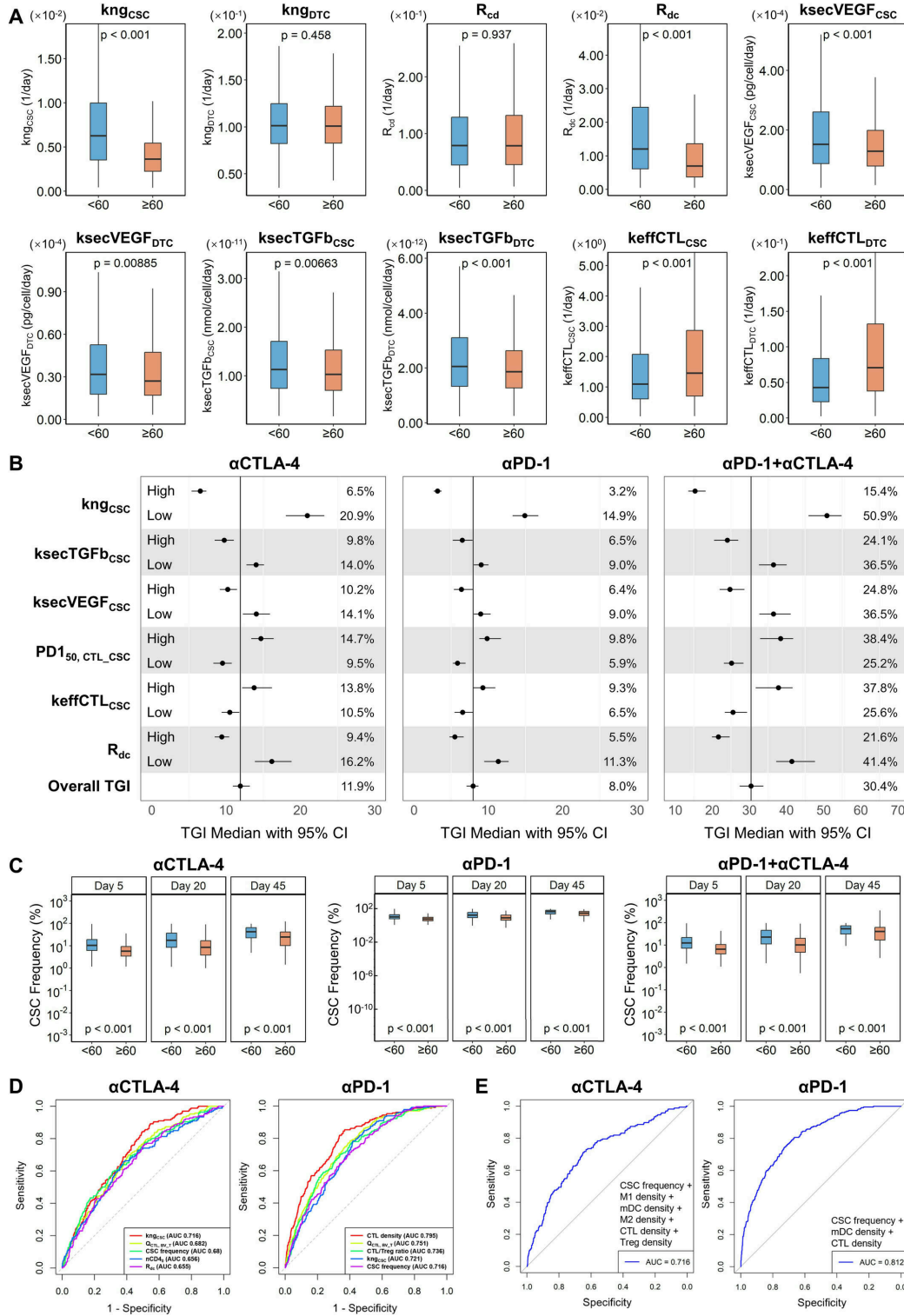


Figure 7: The key model parameters determining ICI efficacy. (A) Comparison of the parameter values of kng_{CSC} , kng_{DTC} , R_{cd} , R_{dc} , $ksecVEGF_{CSC}$, $ksecVEGF_{DTC}$, $ksecTGFb_{CSC}$, $ksecTGFb_{DTC}$, $keffCTL_{CSC}$, $keffCTL_{DTC}$